

Unit 19

NITROGEN COMPOUNDS

26 items · 43 marks

Syllabus sections in this unit

19.1 Primary amines

19.2 Nitriles and hydroxynitriles

Unit 19: Nitrogen compounds**1. M/J 26/P24/Q4/d(ii)**

(d) Butane is converted to pentanenitrile via a two-step synthesis, as shown in Figure 4.3.

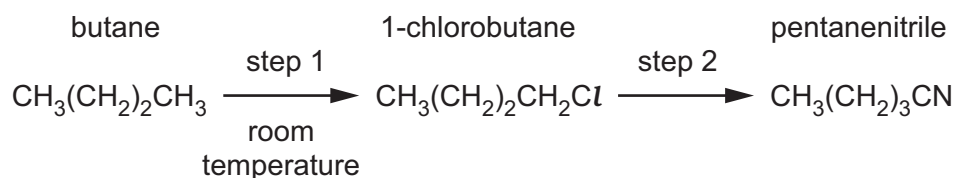


Figure 4.3

(ii) Pentanenitrile reacts with dilute acid.

Draw the displayed formula of the organic product made in this reaction.

[1]

2. MAR 26/P22/Q4/a(iv)

4 Cyanoethanoic acid, $\text{CH}_2(\text{CN})\text{COOH}$, is a weak organic acid.

(a) At room temperature, $\text{CH}_2(\text{CN})\text{COOH}$ undergoes an acid–base reaction with $\text{NaOH}(\text{aq})$.

(iv) Suggest the formula of the organic product when $\text{CH}_2(\text{CN})\text{COOH}(\text{aq})$ is heated under reflux with $\text{NaOH}(\text{aq})$.

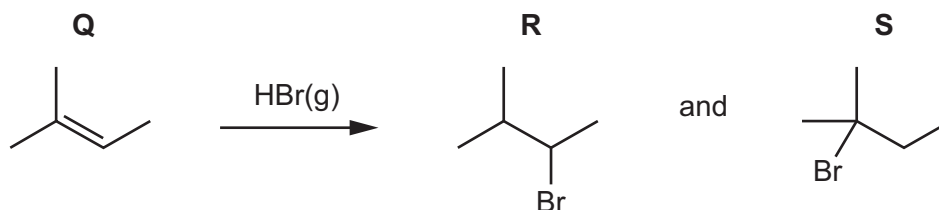
[1]

3. O/N 25/P24/Q4/b(ii)

4 Compounds **P** and **Q** are structural isomers.

**Fig. 4.1**

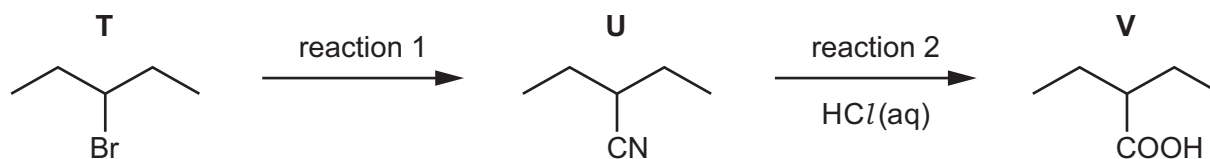
(v) **Q** reacts with HBr(g) to produce two structural isomers, **R** and **S**, as shown in Fig. 4.2.

**Fig. 4.2**

State and explain why isomer **S** is the major product of the reaction.

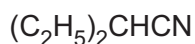
[2]

(b) Fig. 4.3 shows a synthesis starting from **T**, a different isomer of **R** and **S**.

**Fig. 4.3**

(ii) Reaction 2 is a hydrolysis reaction.

Construct an equation for reaction 2.



[1]

4. M/J 25/P22/Q4/b(iii,iv)

(b) Propan-1-ol is converted to compound **T** in a three-step synthesis.

In step 1, SOCl_2 is added to propan-1-ol to produce compound **E**.

In step 2, **E** reacts with a suitable reagent to produce butanenitrile.

In step 3, butanenitrile is heated with NaOH(aq) .

(iii) Construct an equation to describe the reaction of butanenitrile with NaOH(aq) in step 3.

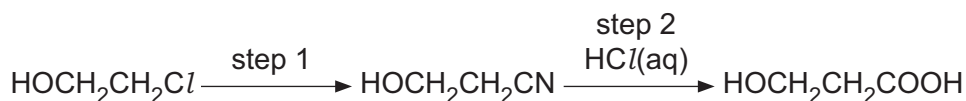
[2]

(iv) Name the type of reaction that occurs in step 3.

[1]

5. M/J 25/P23/Q6/c(ii,iii)

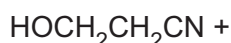
- (c) $\text{HOCH}_2\text{CH}_2\text{Cl}$ reacts in a two-step synthesis to produce $\text{HOCH}_2\text{CH}_2\text{COOH}$, as shown in Fig. 6.2.

**Fig. 6.2**

- (ii) Identify the type of reaction that occurs in step 2.

[1]

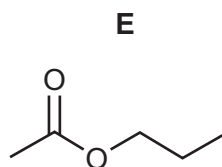
- (iii) Complete the equation to show the reaction in step 2.



[1]

6. M/J 25/P24/Q4/d(i)

- 4 The skeletal formula of **E** is shown in Fig. 4.1.

**Fig. 4.1**

- (d) **F** is an isomer of **E**.

F reacts with HCl(aq) to produce **G** and **H**.

G is a secondary alcohol.

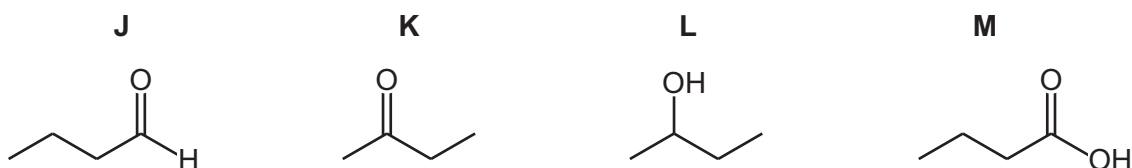
H is also produced when HCl(aq) is added to CH_3CN .

- (i) Construct an equation for the acid hydrolysis of CH_3CN with HCl(aq) .

[1]

7. MAR 25/P22/Q4/b(ii)

- 4 Fig. 4.1 shows compounds **J** to **M**, each of which contains four carbon atoms.

**Fig. 4.1**

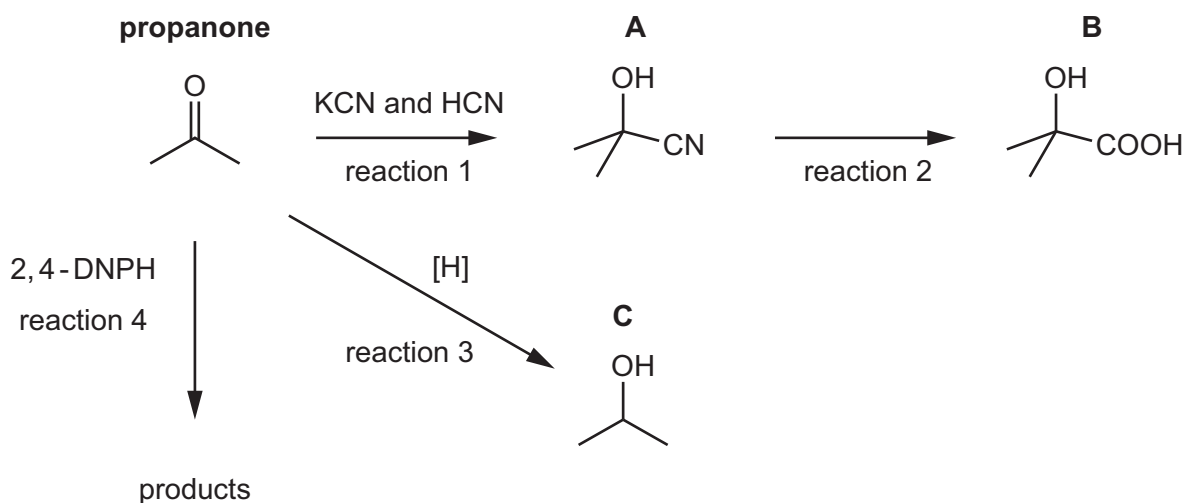
- (b) **K** reacts with HCN in the presence of a KCN catalyst, forming **N**.

- (ii) Identify the class of compound of which **N** is a member.

[1]

8. O/N 24/P21/Q4/b

- 4 Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

- (b) Suggest the reagents and conditions for reaction 2.

[1]

9. O/N 24/P22/Q2/b(ii)

- 2 Oxygen is a Group 16 element.

- (b) H_2O reacts with both inorganic and organic compounds.

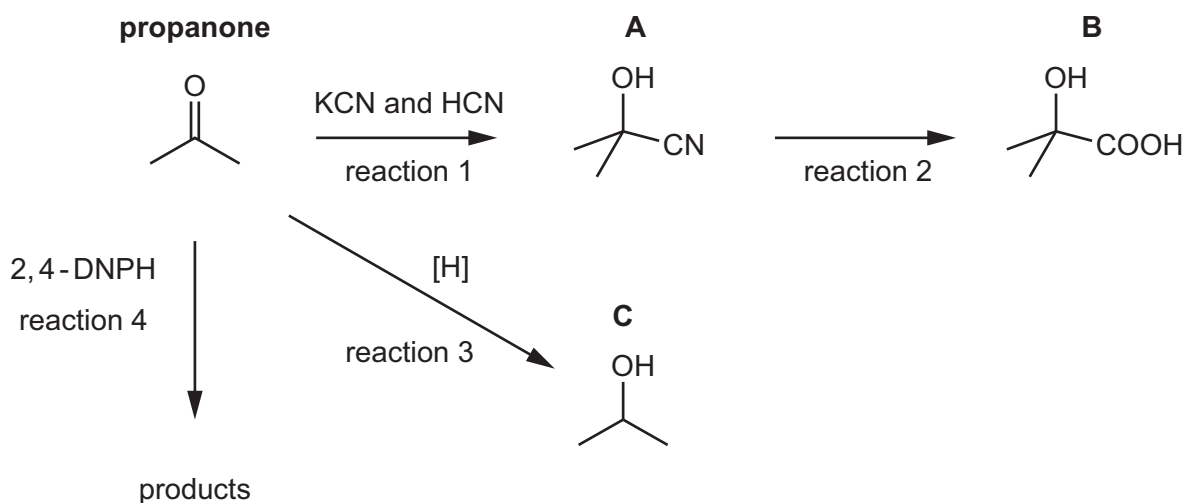
- (ii) Write an equation for the reaction of CH_3CN with H_2O in acidic conditions.



[1]

10. O/N 24/P23/Q4/b

- 4 Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

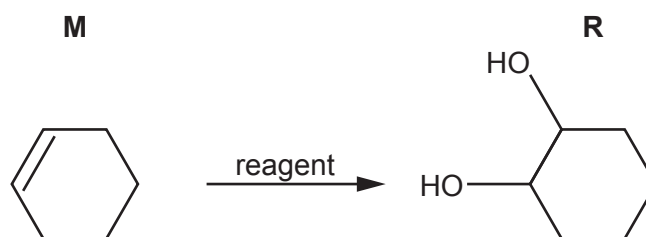
**Fig. 4.1**

- (b) Suggest the reagents and conditions for reaction 2.

[1]

11. M/J 24/P23/Q5/d(i,ii)

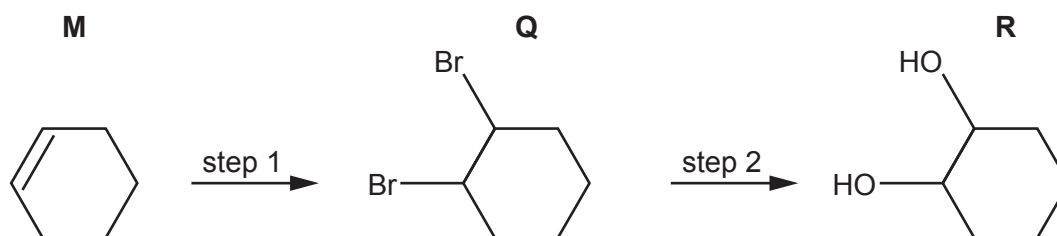
- 5 (a) **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

**Fig. 5.1**

Identify the reagent and conditions for this reaction.

[1]

- (b) **R** is also made from **M** by two steps, as shown in Fig. 5.2.

**Fig. 5.2**

(d) **Y** is made from **Q** in a three-step reaction.

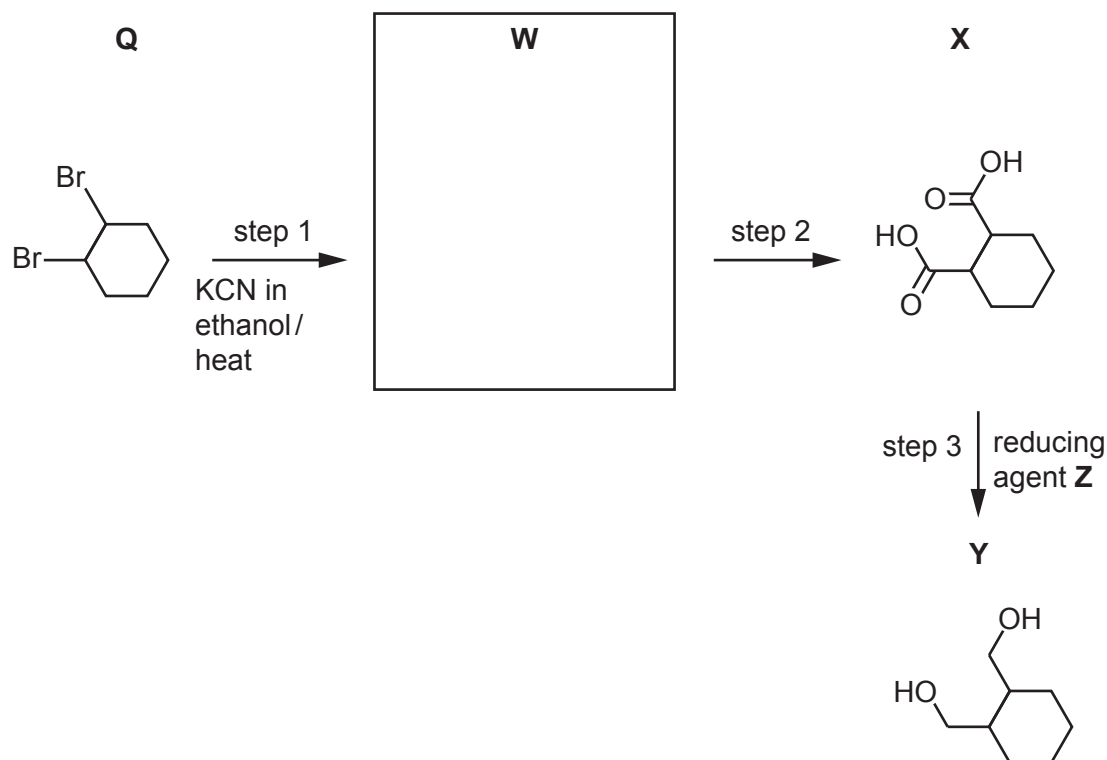


Fig. 5.4

- (i) Draw the structure of **W** in the box in Fig. 5.4. [1]
- (ii) In step 2, **W** is heated with HCl(aq) to produce **X** and an inorganic product. Identify the formula of the inorganic product. [1]

12. MAR 24/P22/Q4/b(iii)

4 Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

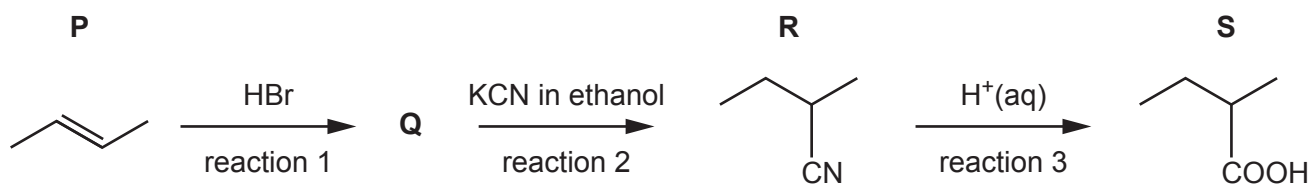
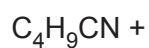


Fig. 4.1

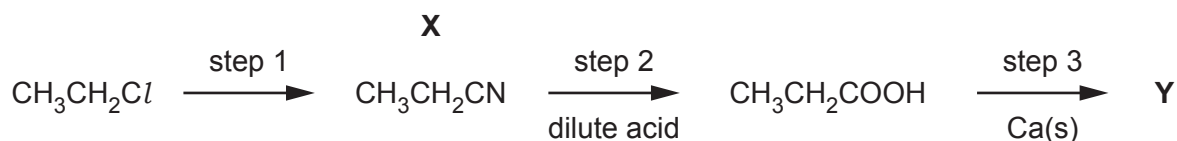
- (iii) Complete the equation for reaction 3. **R** is represented as $\text{C}_4\text{H}_9\text{CN}$.



[1]

13. M/J 23/P22/Q3/a(iii)

- 3 Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from $\text{CH}_3\text{CH}_2\text{Cl}$.

**Fig. 3.1**

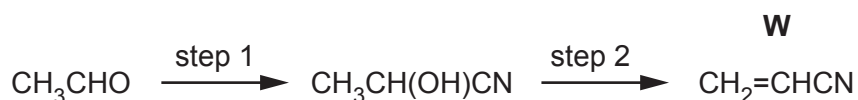
- (iii) Identify the type of reaction that occurs when dilute acid is added to **X** in step 2.

[1]

14. M/J 23/P23/Q6/e,f(ii)

- 6 Compound **W**, $\text{CH}_2=\text{CHCN}$, is used to make an addition polymer which is present in carbon fibres.

- (b) CH_3CHO is used in a two-step synthetic route to form **W**, as shown in Fig. 6.1.
In step 1, CH_3CHO is heated with HCN in the presence of KCN .

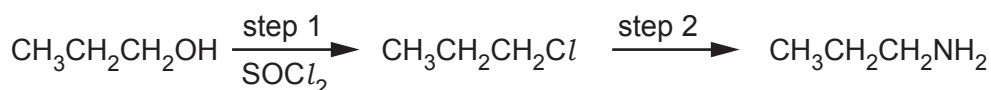
**Fig. 6.1**

- (e) Propanenitrile is heated with hydrogen gas and a platinum catalyst. The only product is propylamine.

Construct an equation for this reaction.

[1]

- (f) Propylamine can also be formed in a two-step synthesis from propan-1-ol, as shown in Fig. 6.4.

**Fig. 6.4**

- (ii) Identify the reagent and conditions for step 2 in Fig. 6.4.

[2]

15. MAR 23/P22/Q4/c(ii)

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

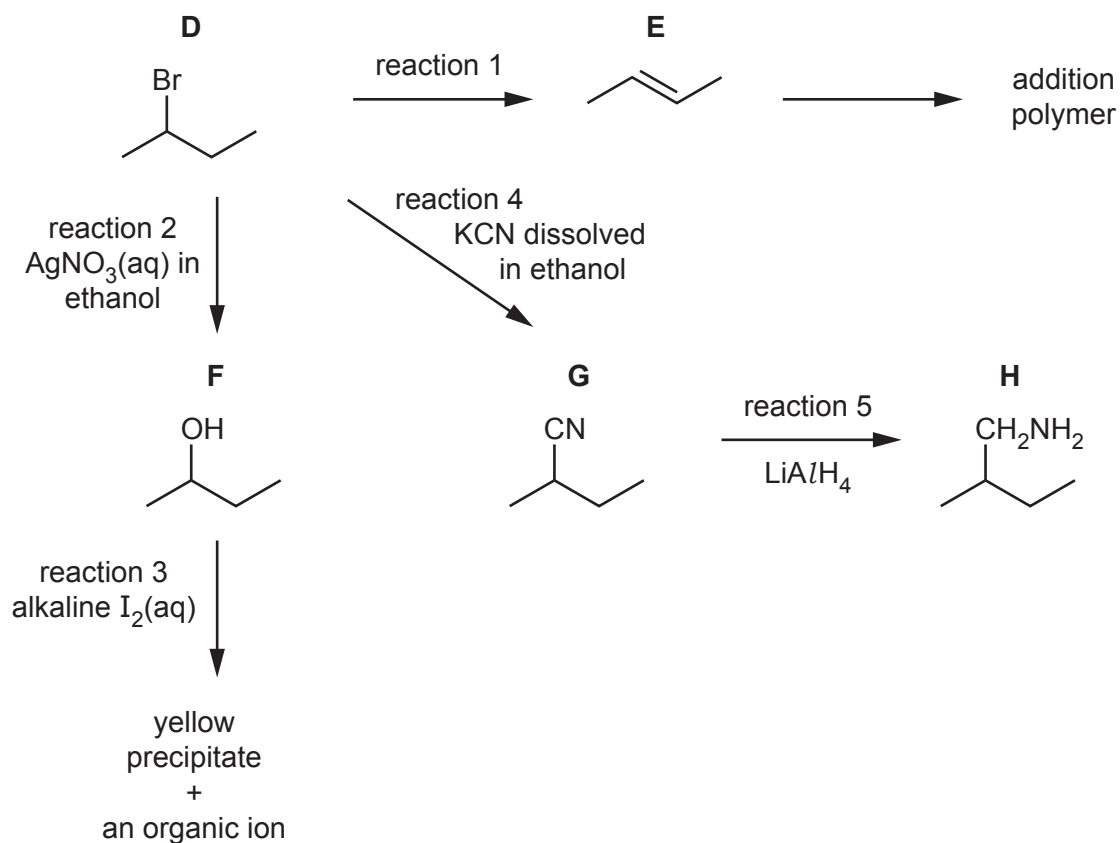


Fig. 4.1

(ii) Reaction 5 is similar to the reaction of LiAlH_4 with carboxylic acids to form alcohols.

Suggest the role of LiAlH_4 in reaction 5.

[1]

16. M/J 22/P22/Q5/c(ii)

5 Fig. 5.1 shows three reactions of 2-bromopropane, $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$.

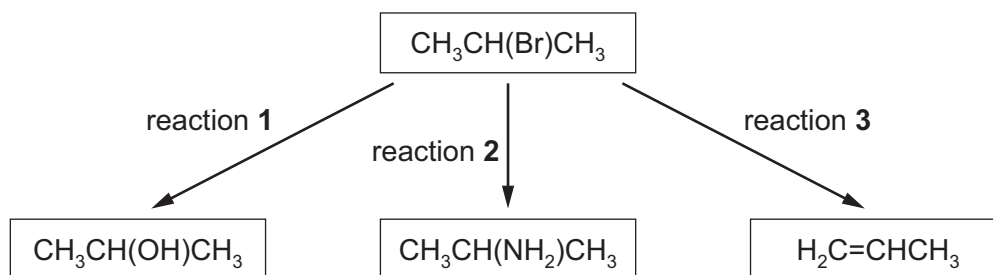


Fig. 5.1

(c) Fig. 5.2 shows how butan-1-ol can be made from 1-bromopropane in three steps.



Fig. 5.2

(ii) In step 2, butanenitrile is heated with $\text{HCl}(\text{aq})$. A hydrolysis reaction occurs.

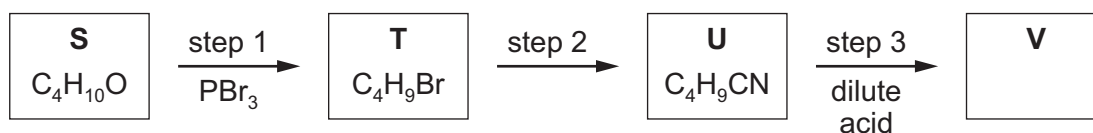
Construct an equation for the reaction in step 2.

[1]

17. M/J 21/P22/Q5/b(iv)

5 **S** is a secondary alcohol with molecular formula $\text{C}_4\text{H}_{10}\text{O}$.

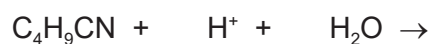
(b) **S** is converted to **V** in a three-step reaction sequence.



In step 1, the secondary alcohol **S** reacts with PBr_3 to produce **T**, which has molecular formula $\text{C}_4\text{H}_9\text{Br}$.

(iv) Step 3 involves heating $\text{C}_4\text{H}_9\text{CN}$ with dilute acid to form **V**.

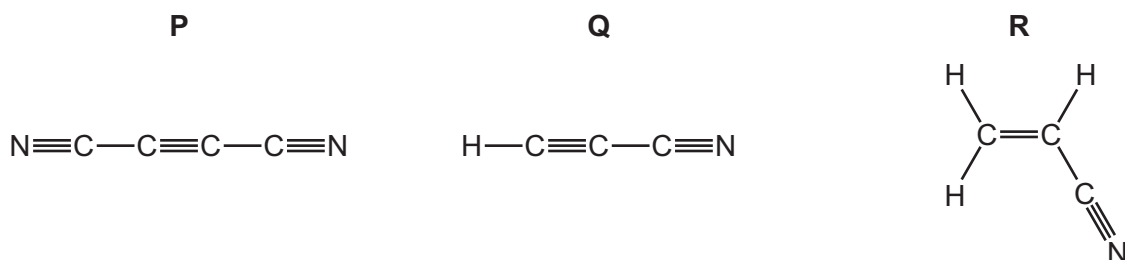
Complete the equation for this reaction.



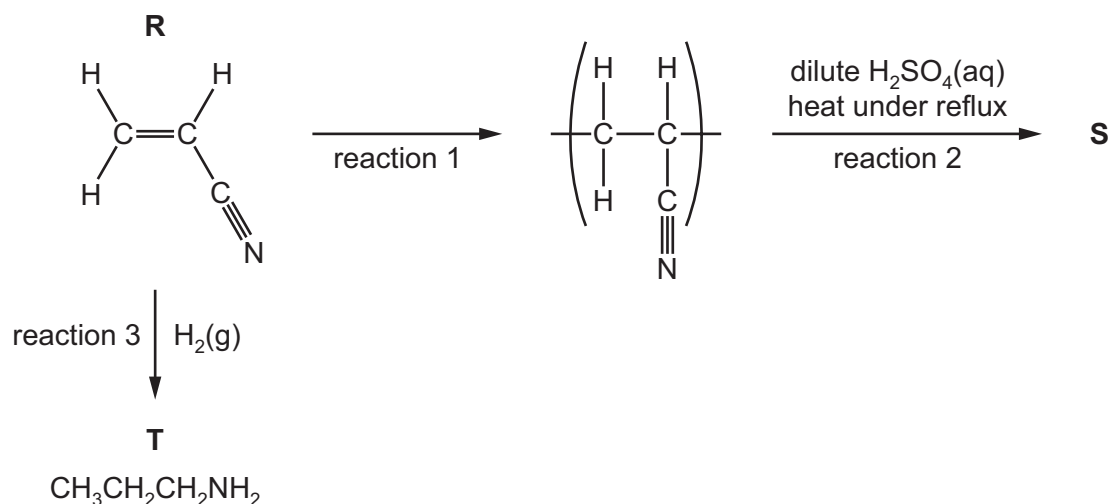
[2]

18. MAR 21/P22/Q3/d

3 Compounds **P**, **Q** and **R** have all been found in the atmosphere of one of Saturn's moons.



(d) The flow chart shows some reactions of **R**.



(i) Name the type of reaction shown in reaction 1.

[1]

(ii) Draw the structure of **S**, the organic product of reaction 2.

[1]

(iii) Name **T**.

[1]

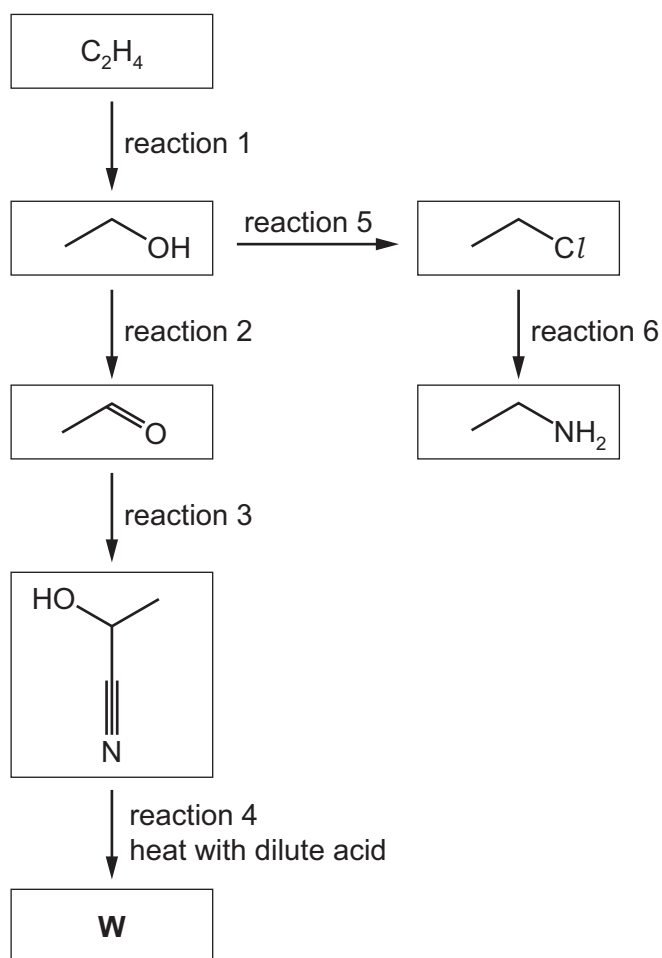
(iv) **T** can also be formed by the reaction of $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ with ammonia.

State the necessary conditions of this reaction.

[1]

19. M/J 20/P22/Q5/b(ii),c

5 The reaction sequence shows how ethene, C_2H_4 , can be converted into other organic molecules.



(b) In reaction 3 the organic molecule reacts with HCN and a KCN catalyst.

(ii) Name the functional groups present in the product of reaction 3.

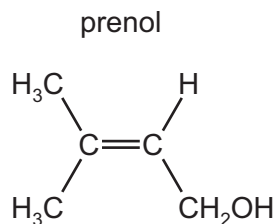
[2]

(c) Draw the structure of the organic molecule **W** formed in reaction 4.

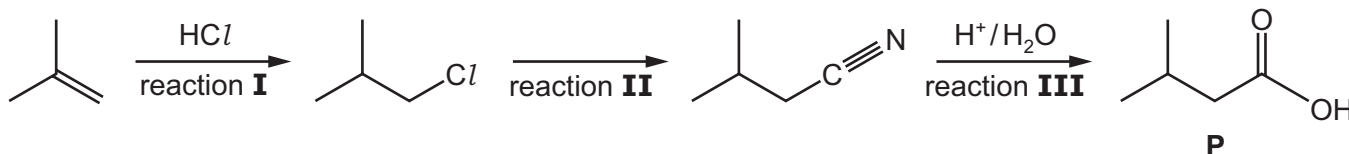
[1]

20. O/N 19/P22/Q4/e(iii)

- 4 Prenol is a naturally occurring organic molecule found in many fruits. It contains both an alkene and an alcohol functional group.



- (e) **P** can be produced as shown.



- (iii) Name the type of reaction that occurs in reaction **III**.

[1]

21. M/J 19/P21/Q5/c

- 5 Ethanal reacts with a mixture of HCN and NaCN to make 2-hydroxypropanenitrile, $\text{CH}_3\text{CH}(\text{OH})\text{CN}$.

The reaction mechanism is nucleophilic addition.

- (c) Give the structure of the organic product of the reaction of $\text{CH}_3\text{CH}(\text{OH})\text{CN}$ with dilute sulfuric acid.

[1]

22. M/J 19/P23/Q5/a(i),b

- 5 Halogenoalkanes react with a number of different reagents in nucleophilic substitution reactions.

- (a) A sample of potassium cyanide dissolved in ethanol is added to a sample of 1-bromobutane, $\text{CH}_3(\text{CH}_2)_3\text{Br}$, and heated under reflux. A nucleophilic substitution reaction occurs and compound **A** is formed.

- (i) Name compound **A**.

[1]

- (b) State the reagent(s) and conditions needed for $\text{CH}_3(\text{CH}_2)_3\text{Br}$ to react to form $\text{CH}_3(\text{CH}_2)_3\text{NH}_2$.

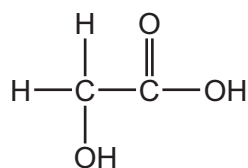
reagent(s)

conditions

[2]

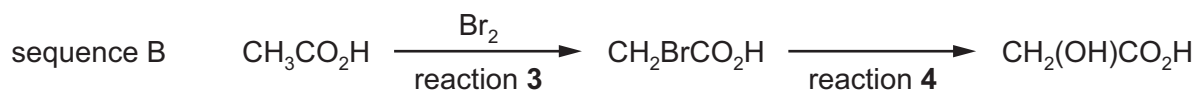
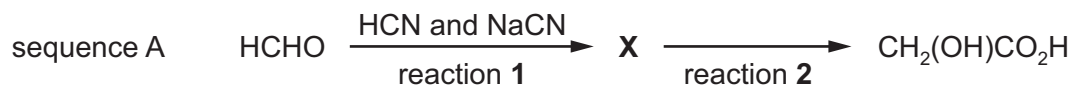
23. O/N 18/P21/Q4/b(i,ii)

4 The structure of glycolic acid is shown.



glycolic acid

(b) Two reaction sequences to make glycolic acid are shown.



(i) Draw the structure of **X**.

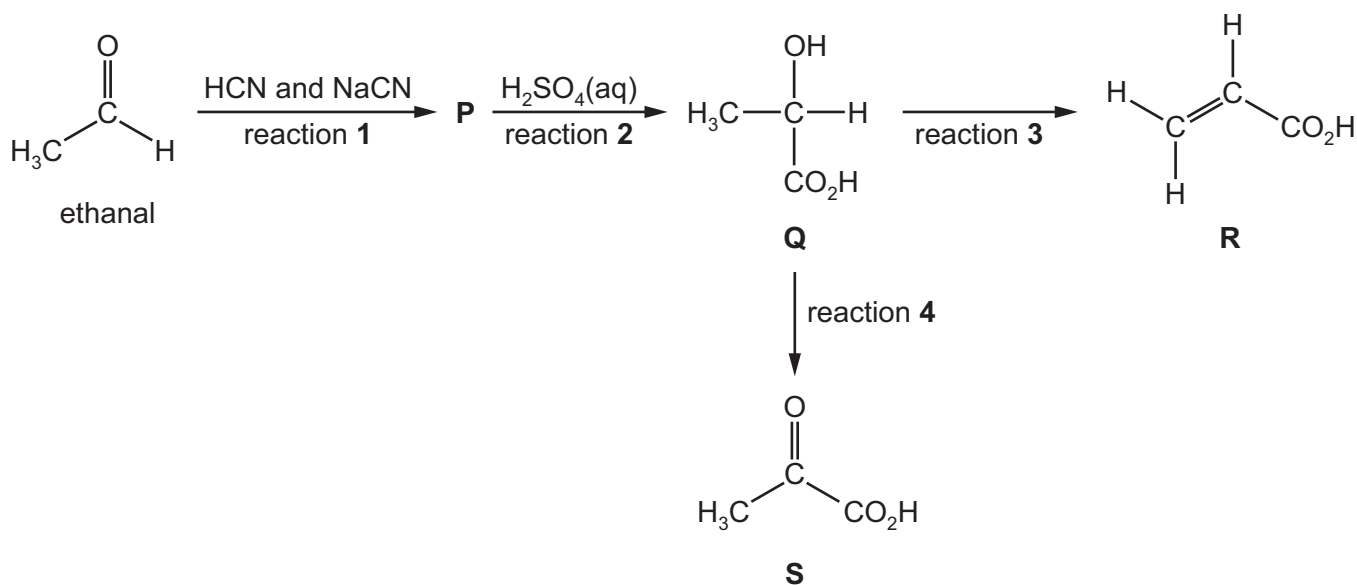
[1]

(ii) Name the reagent for reaction 2.

[1]

24. O/N 18/P22/Q4/a(i,ii)

4 The diagram shows a reaction sequence starting from ethanal.



(a) (i) Draw the **displayed** formula of **P**.

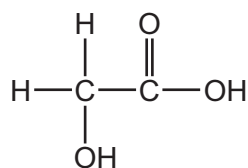
[1]

(ii) Name the type of chemical reaction that occurs in reaction 3.

[1]

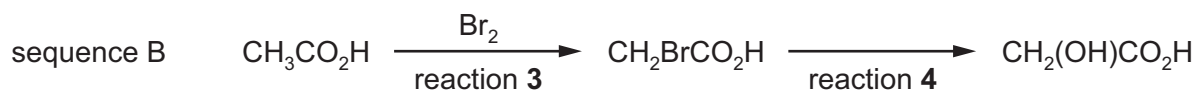
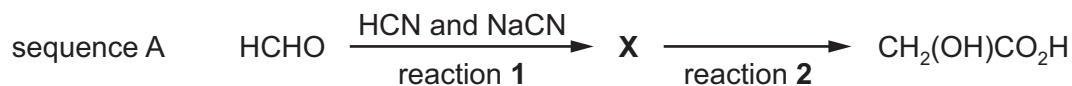
25. O/N 18/P23/Q4/b(i,ii)

4 The structure of glycolic acid is shown.



glycolic acid

(b) Two reaction sequences to make glycolic acid are shown.



(i) Draw the structure of X.

[1]

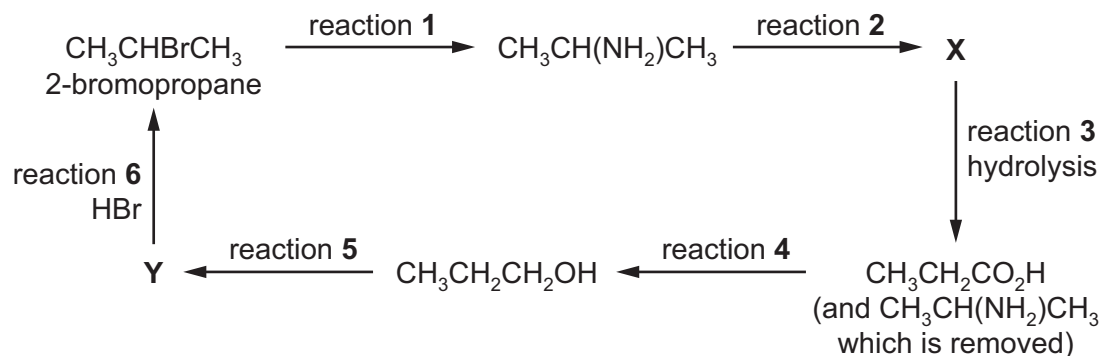
(ii) Name the reagent for reaction 2.

[1]

26. O/N 15/P22/Q4/b(i)

4 Halogenoalkanes are useful intermediates in the synthesis of a wide variety of compounds.

(b) Some reactions involving 2-bromopropane are shown.



(i) State the reagent needed for reaction 1.

[1]

Answers Section

1. M/J 26/P24/Q4/d(ii)

4(d)(ii)	$ \begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{H} & & \text{H} & & \text{O} \\ & & & & & & & & & // \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C} \\ & & & & & & & & & \backslash \\ & \text{H} & & \text{H} & & \text{H} & & \text{H} & & \text{O} - \text{H} \end{array} $	1	R-OH
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2. MAR 26/P22/Q4/a(iv)

4(a)(iv)	CH ₂ (COO) ₂ Na ₂ OR CH ₂ (COONa) ₂	1
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3. O/N 25/P24/Q4/b(ii)

4(b)(ii)	(C ₂ H ₅) ₂ CHCN + 2H ₂ O + HCl → (C ₂ H ₅) ₂ CHCOOH + NH ₄ Cl	1
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4. M/J 25/P22/Q4/b(iii,iv)

4(b)(iii)	CH ₃ CH ₂ CH ₂ CN + NaOH + H ₂ O → CH ₃ CH ₂ CH ₂ CO ₂ Na + NH ₃ OR CH ₃ CH ₂ CH ₂ CN + NaOH + 2H ₂ O → CH ₃ CH ₂ CH ₂ CO ₂ Na + NH ₄ OH M1 correct organic product (as molecule or anion) M2 rest of equation is correct	2
4(b)(iv)	(alkaline) hydrolysis	1

5. M/J 25/P23/Q6/c(ii,iii)

6(c)(ii)	(acid) hydrolysis	1
6(c)(iii)	HOCH ₂ CH ₂ CN + 2H ₂ O + HCl → HOCH ₂ CH ₂ COOH + NH ₄ Cl	1

6. M/J 25/P24/Q4/d(i)

4(d)(i)	CH ₃ CN + HCl + 2H ₂ O → CH ₃ COOH + NH ₄ Cl	1
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7. MAR 25/P22/Q4/b(ii)

4(b)(ii)	hydroxynitrile	1
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8. O/N 24/P21/Q4/b

4(b)	H ₂ SO ₄ (aq) or HCl(aq)	1
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9. O/N 24/P22/Q2/b(ii)

2(b)(ii)	CH ₃ CN + 2H ₂ O + (1)H ⁺ → CH ₃ COOH + NH ₄ ⁺	1
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10. O/N 24/P23/Q4/b

4(b)	H ₂ SO ₄ (aq) or HCl(aq)	1
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11. M/J 24/P23/Q5/d(i,ii)

5(d)(i)	W =	1
5(d)(ii)	NH ₄ Cl	1

12. MAR 24/P22/Q4/b(iii)

4(b)(iii)	$\text{C}_4\text{H}_9\text{CN} + 2\text{H}_2\text{O} + \text{H}^+ \rightarrow \text{C}_4\text{H}_9\text{COOH} + \text{NH}_4^+$	1
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13. M/J 23/P22/Q3/a(iii)

3(a)(iii)	(acid) hydrolysis	1
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14. M/J 23/P23/Q6/e,f(ii)

6(e)	$\text{CH}_3\text{CH}_2\text{CN} + 2\text{H}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$	1
6(f)(ii)	M1 ammonia / NH_3 M2 heat + (M1 in ethanol) under pressure	2

15. MAR 23/P22/Q4/c(ii)

4(c)(ii)	reducing agent	1
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16. M/J 22/P22/Q5/c(ii)

5(c)(ii)	$\text{CN}(\text{CH}_2)_2\text{CH}_3 + 2\text{H}_2\text{O} + \text{HCl} \rightarrow \text{HOOC}(\text{CH}_2)_2\text{CH}_3 + \text{NH}_4\text{Cl}$ OR 	1
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17. M/J 21/P22/Q5/b(iv)

5(b)(iv)	$(1)\text{C}_4\text{H}_9\text{CN} + (1)\text{H}^+ + 2\text{H}_2\text{O} \rightarrow (1)\text{C}_4\text{H}_9\text{CO}_2\text{H} + (1)\text{NH}_4^+ \checkmark\checkmark$ <i>correct organic product showing carboxylic acid functional group</i> M1 $\text{C}_4\text{H}_9\text{CO}_2\text{H}$ / $\text{C}_4\text{H}_9\text{COOH}$	1
	M2 balanced equation with a $\text{C}_5\text{H}_{10}\text{O}_2$ or equivalent structure as product	1

18. MAR 21/P22/Q3/d

3(d)(i)	addition polymerisation	1
3(d)(ii)		1
3(d)(iii)	propan-1-amine / 1-aminopropane	1
3(d)(iv)	alcoholic / ethanolic solution AND high pressure / heat in a sealed container	1

19. M/J 20/P22/Q5/b(ii),c

5(b)(ii)	ALLOW in any order M1 nitrile M2 (secondary) alcohol	2
5(b)(iii)	 ALLOW any unambiguous structure	1

20. O/N 19/P22/Q4/e(iii)

4(e)(iii)	(acidic) hydrolysis	1
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21. M/J 19/P21/Q5/c

5(c)	$\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{H}$ OR $\text{HO}_2\text{CCH}(\text{OH})\text{CH}_3$	1
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22. M/J 19/P23/Q5/a(i),b

5(a)(i)	pentanenitrile	1
5(b)	M1 <i>reagent</i> Ammonia M2 <i>conditions</i> heat with under pressure / heat in a sealed tube	2

23. O/N 18/P21/Q4/b(i,ii)

4(b)(i)	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{C}\equiv\text{N} \\ \\ \text{OH} \end{array}$	1
4(b)(ii)	hydrochloric / sulfuric / nitric / phosphoric acid	1

24. O/N 18/P22/Q4/a(i,ii)

4(a)(i)	$\begin{array}{c} & & \text{H} \\ & & \\ \text{H} & -\text{C} & -\text{C}-\text{O}-\text{H} \\ & & \\ & \text{H} & \text{C} \\ & & \\ & & \text{N} \end{array}$	1
4(a)(ii)	dehydration	1

25. O/N 18/P23/Q4/b(i,ii)

4(b)(i)	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{C}\equiv\text{N} \\ \\ \text{OH} \end{array}$	1
4(b)(ii)	hydrochloric / sulfuric / nitric / phosphoric acid	1

26. O/N 15/P22/Q4/b(i)

(b) (i)	ammonia / NH_3	[1]	[1]
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